RELATIVE APPROXIMATION OF MORPHISMS OF FINITE TYPE

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Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Limits and fibre products. Limits, limits commute lemmas of the chapter

Source reference. Original lemma 0GS1: sources/stacks-chow/tags.txt, source lines 20916–20916. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 002M: sources/stacks-chow/tags.txt, source lines 103–103. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01KR: sources/stacks-chow/tags.txt, source lines 2060–2060. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01SD: sources/stacks-chow/tags.txt, source lines 2334–2334. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01SC: sources/stacks-chow/tags.txt, source lines 2333–2333. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01TX: sources/stacks-chow/tags.txt, source lines 2390–2390. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02FV: sources/stacks-chow/tags.txt, source lines 3206–3206. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01TS: sources/stacks-chow/tags.txt, source lines 2385–2385. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Assume that*

(1) *X is quasi-compact and quasi-separated, and*
(2) *S is quasi-separated.*

Then $X = \lim X_i$ is a limit of a directed system of schemes X_i of finite presentation over S with affine transition morphisms over S.

Proof. Since $f(X)$ is quasi-compact we may replace S by a quasi-compact open containing $f(X)$. Hence we may assume S is quasi-compact. By Lemma 0GS1 we can write $(X \to S) = \lim(X_i \to S_i)$ for some directed inverse system of morphisms

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of finite type schemes over $\mathbf{Z}$ with affine transition morphisms. Since limits commute with limits (Categories, Lemma 002M) we have $X = \lim X_i \times_{S_i} S$. Let $i \geq i'$ in I. The morphism $X_i \times_{S_i} S \to X_{i'} \times_{S_{i'}} S$ is affine as the composition

$$X_i \times_{S_i} S \to X_i \times_{S_{i'}} S \to X_{i'} \times_{S_{i'}} S$$

where the first morphism is a closed immersion (by Schemes, Lemma 01KR) and the second is a base change of an affine morphism (Morphisms, Lemma 01SD) and the composition of affine morphisms is affine (Morphisms, Lemma 01SC). The morphisms f_i are of finite presentation (Morphisms, Lemmas 01TX and 02FV) and hence the base changes $X_i \times_{f_i, S_i} S \to S$ are of finite presentation (Morphisms, Lemma 01TS). $\square$